

Ing. Michal Štěpán

Senior GenAI Engineer · AI Solution Architecture · RAG & Agents · Python/FastAPI · Azure

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📍 Sviadnov (Ostrava), Czechia · remote-first / EU

📅 Open to B2B / contract consulting

🌐 mxstex.github.io/MyWebPage

📄 Czech native · English B2-C1 (Cambridge FCE)

Profile

Senior GenAI Engineer focused on turning ambiguous AI use cases into production systems. I work end to end across solution architecture, backend engineering, evaluation, cloud delivery and operations: RAG and document intelligence, agent-based assistants, structured LLM outputs with deterministic validation, golden-dataset evaluation and FastAPI services on Azure. I bring 15+ years across industrial software, data, cloud and energy, including previous freelance delivery and technical leadership. I am strongest where AI must be reliable, testable and integrated into real business workflows. Open to B2B / contract work in GenAI engineering, AI solution architecture and production AI delivery.

AI solution architecture

use case → design → PoC → production

Production GenAI

RAG · agents · Document AI · FastAPI

Quality & evaluation

golden datasets · deterministic checks · stress tests

Cloud delivery

Azure · GCP · Docker · Terraform · CI/CD

Core skills

● GenAI & LLM applications

Azure OpenAI / AI Foundry · Azure AI Agents
RAG · Azure AI Search · ChromaDB · Document Intelligence
structured outputs · LLM + deterministic rules
golden datasets & evaluation · MCP · FastMCP
LLM agents on Ollama · pgvector memory · Claude Code
GitHub Copilot

● Cloud & delivery

Azure Container Apps · Container Registry
Key Vault · managed identity · Log Analytics
Blob & Table Storage · Azure DevOps / Pipelines
GCP Cloud Run · Secret Manager · Terraform (GCP) · Docker
OAuth 2.0 / RS256 JWT

● Programming & testing

Python · FastAPI · Pandas · Pytest · SQL / PostgreSQL
Redis · Neo4j · NiceGUI · Vue 3 · Playwright
load & stress tests · NumPy / SciPy

● Worked with earlier

LangChain · LangGraph · Dagster · Airflow · DuckDB
BigQuery · RabbitMQ · Kubernetes · GitLab CI
GitHub Actions · Flask · Streamlit · AWS S3 · Lambda
Azure Functions · Grafana · Power BI · Kotlin · WebGL2
ARIMA / SARIMA

Professional experience

● Senior GenAI Application Engineer · ABB s.r.o.

Sep 2025 – present

- Design and build production GenAI solutions for industrial engineering workflows, from use-case analysis and architecture through implementation, evaluation and cloud delivery.
- Implement document extraction and RAG, structured LLM outputs with deterministic validation, and agent-based assistants using Azure OpenAI / AI Foundry and Azure AI Agents.
- Own AI quality and evaluation: golden datasets, acceptance criteria, deterministic checks, performance and stress tests, and cloud-vs-local model comparison through Ollama.
- Develop Python/FastAPI services and integrate PostgreSQL, Redis and Neo4j where relational, session/cache and graph data are required.
- Deliver across dev, test, stage and prod using Azure Pipelines, Docker, Container Registry, Azure Container Apps, Key Vault with managed identity and Log Analytics; built an MCP/FastMCP PoC exposing Azure DevOps work items to coding assistants through controlled tools.

STACK Python · FastAPI · Azure OpenAI / AI Foundry · Azure AI Agents · RAG · ChromaDB · Document Intelligence · PostgreSQL · Redis · Neo4j · Pandas · Pytest · Docker · Azure Container Apps · Key Vault · Azure Pipelines · MCP · Ollama

Freelance Software / Data Engineer

Sep 2022 – Sep 2025

B2B delivery for DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing · remote, international teams

- Delivered backend and data-engineering work directly into client environments, taking features from requirements through implementation, testing and production support.
- Designed end-to-end Python data workflows with Dagster and Airflow using PostgreSQL, DuckDB and BigQuery; built REST services and RabbitMQ-based high-volume processing.
- Migrated services from Kubernetes to cloud-native GCP services including Cloud Run, GCS and BigQuery.
- Maintained CI/CD in GitLab CI and GitHub Actions with Pytest-based unit, integration and end-to-end testing.
- Optimised high-performance processing pipelines for recommender systems and provided L2 engineering support for production systems.

STACK Python · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · GCP · Kubernetes · GitLab CI · GitHub Actions

Senior Software Developer · Siemens s.r.o.

Jan 2021 – Aug 2022

- Developed an Energy Management System consolidating and analysing data from many sources.
- Implemented ARIMA and SARIMA models for time-series forecasting.
- Mentored junior developers.

STACK Python · Flask · FastAPI · Docker · PostgreSQL · Grafana · Power BI · RabbitMQ · AWS S3 · AWS Lambda

Software Project Engineer · ABB s.r.o.

Oct 2018 – Jan 2021

- Developed a water and energy consumption evaluation system; designed and implemented predictive-analytics features.

STACK Python · Docker · PostgreSQL · Azure Functions · Azure Blob Storage · Grafana · Power BI

IT Specialist · Power Engineer · Technical Deputy · DISTEP a.s.

May 2010 – Oct 2018

- As technical deputy, led an operations team of more than fifteen people in water and heat distribution.
- Improved resource efficiency through optimised calculations and monitoring; streamlined data collection with PDA barcode readers (MS SQL, VBA).

Independent engineering portfolio · Tojin Games

Live browser products, public YouTube demos and pre-release engineering projects · youtube.com/@tojin111 · playable builds on the portfolio site

- **Production science software:** live Cloud Run projects include Gravity 3D, Orbit, Ion Drive, Starforge, MyZoo, My Garden, Reactor Operator and Quantum. Ion Drive models N-body trajectories, electric/nuclear propulsion and atmospheric scooping; Reactor Operator models reactor kinetics, delayed neutrons, xenon and thermal feedback across seven reactor designs. More than 80 public demos, trailers and explainers on YouTube.
- **GenAI & autonomous agents:** Jednota uses local LLM strategist, diplomat and fleet agents with structured outputs, deterministic validation and pgvector memory; its Jednota: Tactics battle simulator is live on Cloud Run and an online multiplayer version (FastAPI, PostgreSQL, Vue 3) is in development. Tojin Forge (development preview on YouTube) is a local multi-repository coding agent: Ollama tool loop, sandboxed write scopes, checkpoints, CLI/MCP, a GPU-aware model picker and unattended overnight runs gated by each repository's own tests.
- **Pre-release engineering:** Particle Forge teaches the Standard Model through simulated experiments across a complete tested campaign; browser, persistence and Docker validation are complete while public video/cloud release work continues.
- **Platform & automation:** shared FastAPI authentication/entitlements use OAuth 2.0, one-time codes and short-lived RS256 JWTs; services run on GCP with Secret Manager and Terraform. A code-driven pipeline records real gameplay in 4K through Playwright/CDP and ffmpeg/NVENC, with local TTS/ComfyUI and automated output validation.

Education & certificates

Ing. (MSc), Electrical Engineering and Computer Science · 2012

VŠB – Technical University of Ostrava. Thesis: VBA Utilization for Power Engineering Data Processing and Presentation

Bc., Nuclear Engineering · 2009

Czech Technical University in Prague, Faculty of Nuclear Sciences and Physical Engineering. Thesis: Physical Analysis of Fast Reactor Cores in Up-to-Date Tendencies

Research project · 2006

Institute of Plasma Physics, Czech Academy of Sciences: electrostatic turbulence in the edge plasma of the CASTOR tokamak

Microsoft AZ-305 · AZ-400

Azure infrastructure design and DevOps solutions · training

Cambridge English: First (FCE)

Grade B · CEFR B2 · 2019